

Complete solutions to all problems are required.

Motivations for all theoretical reasonings and for conclusions must be given, you must name the theorems that you might use. All calculations must be possible to follow in your solutions. Intermediate calculations in your solutions must be given.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Red pencil or pen is not allowed.

Pocket calculators are allowed (without internet connection). Only statistical tables provided by the institute are allowed. No other tables or formula sheets are allowed.

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1. Let X be a random variable with probability density function

$$f(x) = \begin{cases} c, & 0 \leq x \leq 1, \\ c(x-1), & 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Determine c so that f is a probability density function. [5p]
(b) Derive and plot the distribution function of X . [5p]

2. Let $X_1, \dots, X_n$ be independent identically distributed random variables that can take two possible values, $X_i = 0$ with probability $1/4$ and $X_i = 1$ with probability $3/4$, for $i = 1, \dots, n$. Let Y be a random variable that is independent of $X_1, \dots, X_n$ and suppose that Y is $\text{Bin}(m, 3/4)$ distributed.

- (a) Define the random variable

$$Z = X_1 + \dots + X_n + Y.$$

Find the probability mass function of Z . [4p]

- (b) Suppose that $m = 4$. Calculate $P(Y \geq X_1 + X_2)$. [6p]

3. Let X be a random variable with density

$$f(x) = \begin{cases} \frac{1}{2}e^{-\lambda}\lambda^x/x! & \text{if } x = 0, 1, 2, 3, \dots, \\ \frac{1}{2}ae^{a(x+1)} & \text{if } x \leq -1, \end{cases}$$

with respect to the measure $\mu(x) = \mu_1(x)1\{x \leq -1/2\} + \mu_2(x)1\{x > -1/2\}$, with μ_1 length measure and μ_2 counting measure on the integers. Suppose that $\lambda = 2$ and $a = 1/3$.

- (a) Calculate $P(-2 \leq X \leq 2)$ and $P(-2 \leq X | X \leq 1)$. [4p]

- (b) Determine and plot the function G defined as

$$G(u) = P(X \geq u | X \leq -1),$$

for $u \leq 0$. Is G a distribution function? [6p]

4. A manufacturing plant P is producing items I . The quality of the items produced depends on whether the items are produced before lunch, after lunch or on weekends. Items produced before lunch have a reliability of 99.9%, meaning that 1 out of 1000 on average breaks when put in use. Items produced after lunch have a reliability of 99.0% and items produced on weekends have a reliability of 95.0%. The proportions of the items that are produced at the plant are 40 % before lunch, 30 % after lunch and the rest on weekends.

- (a) What is the probability that an item produced at the plant breaks when put into use? [10p]

- (b) What is the probability that an item that does break when put into use has been produced on a weekend? [10p]

5. Suppose we are given a probability space $(\Omega, \mathcal{F}, P)$, where $\Omega = [0, 1]$ is the outcome space, $\mathcal{F}$ is the σ -algebra generated by the open intervals in Ω , and P is the probability defined by $P((a, b)) = b - a$ for any $0 \leq a < b \leq 1$. Suppose that $A_1, A_2, \dots$ is a sequence of events, defined as the half-open intervals $A_k = [1/2, 1/2 + 1/k)$, for $k = 1, 2, \dots$. Let

$$B_n = \bigcap_{k=1}^n A_k$$

- (a) Show that $B_n \downarrow A$ as $n \rightarrow \infty$, where A is the one-point set $A = \{1/2\}$. [10p]

- (b) Determine $P(\{1/2\})$. [10p]

6. Suppose that $U_1, U_2, \dots$ is an infinite sequence of independent identically distributed random variables and that all are $Un([9, 11])$ distributed. Define the partial products

$$\begin{aligned} Q_n &= \prod_{i=1}^n U_i \\ &= U_1 \cdot \dots \cdot U_n, \end{aligned}$$

for every $n \geq 1$.

- (a) Find (some appropriate) sequence of real numbers $a_n \uparrow \infty$ such that $P(Q_n > a_n) = 1$ for every n . Show that

$$Q_n \xrightarrow{a.s.} \infty,$$

as $n \rightarrow \infty$. [10p]

- (b) Show that

$$(Q_n)^{1/n} \xrightarrow{P} c,$$

as $n \rightarrow \infty$, for some constant c , and determine the constant c . You may present c using an analytic expression for the numerical value of c . [10p]